

Spray Quenching

Introduction

Adapted from “Spray Quenching” in *Handbook of Quenchants and Quenching Technology* (Ref 1)

The term *spray quenching* refers to a wide variety of quenching processes that involve heat removal facilitated by the impingement of a quenchant medium on a hot metal surface. Some of these processes have obvious differences, while others are similar and differ only in degree. Examples include the addition of droplets of water (or other volatile liquids) to a gas quenching stream in fog quenching (Ref 1); quenching with water or water/air streams (Ref 2); sprays of volatile liquid quenchants other than water (Ref 3); and high-pressure jets of an oil (Ref 4), water (Ref 5, 6–7), or aqueous polymer solution (Ref 8) under the liquid level in a bath.

One advantage of spray quenching, relative to other quench methods, is that a large and adjustable range of cooling rates is achievable by simple changes in flow rates and pressures. The high rates of heat extraction possible with sprays are critical for attaining a good depth of hardness. Segerberg (Ref 5) has examined

this and illustrated the results with the cooling curves shown in Fig. 1. Immersion quenching was performed using an oil and a polymer solution. The water flux for the spray quench was chosen to give a cooling curve with a vapor film of the same duration as that for the oil immersion quench. The quantitative impact of the higher rate of cooling found with the spray system was determined using 35 by 100 mm (1.4 by 4.0 in.) cylindrical samples of SAE 52100 ball bearing steel. These cylinders were heated to 850 °C (1560 °F) and quenched in a fast oil at 50 °C (120 °F), or a 15% aqueous solution of a polyalkylene glycol (PAG) polymer at 25 °C (77 °F), or by the water spray quenching system. Hardness distributions along the diameter of the cylinder are shown in Fig. 2. The higher cooling rates achieved by the immersion quench in the PAG solution and by the water spray quench account for the greater depth of hardness shown in Fig. 2.

Spray quenching is used to optimize heat transfer for hardening while simultaneously developing the desired distribution and level of stress (Ref 9). With spray quenching, the heat-transfer coefficient from the part to the quenching medium is directly related to the flow rate, turbulence, and impingement pressures of the quenchant at the hot surface. It is possible to adjust these parameters during the quench to yield a cooling profile achievable in no other way (Ref 2, 5, 10,11–12). Computerized adjustment of these variables during the quench also have been extensively explored (Ref 2, 5, 10, 13,14,15,16,17–18). This ability to control and vary the heat-transfer coefficient

during the quench may produce properties that would otherwise require more expensive, higher-alloy materials (Ref 9, 19–22).

The next section of this article, “Water Quenching Heat Transfer,” reviews heat-transfer characteristics of immersion quenching and spray quenching with water. Liquid quenching provides high cooling rates required for the steel hardening process. Water is the most commonly used liquid in quenching because it is readily available, easy to pump, nontoxic, and inflammable. Quenching can be carried out by directly submerging the part in a quench bath, bottom reflooding, falling film, sprays, and impinging jets. Impinging jets are used to remove extensive amounts of heat locally, while sprays are used to achieve less but more uniformly distributed cooling rates.

Concepts of Spray Quenching. When quenching a hot metal part in a water bath, the cooling mechanisms vary according to regions, as illustrated in Fig. 3 (Ref 13, 23,

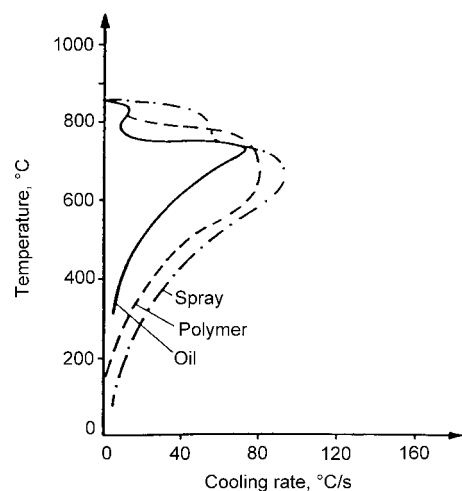


Fig. 1 Cooling rates for spray, polymer, and oil immersion quenching

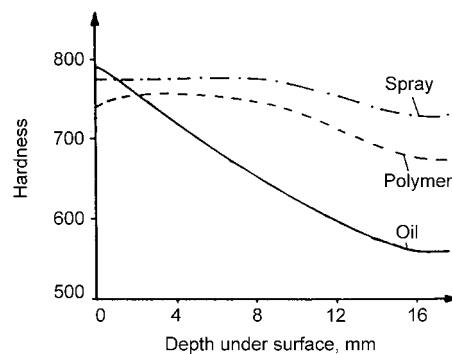


Fig. 2 Hardness distributions for spray, polymer, and oil immersion quenching

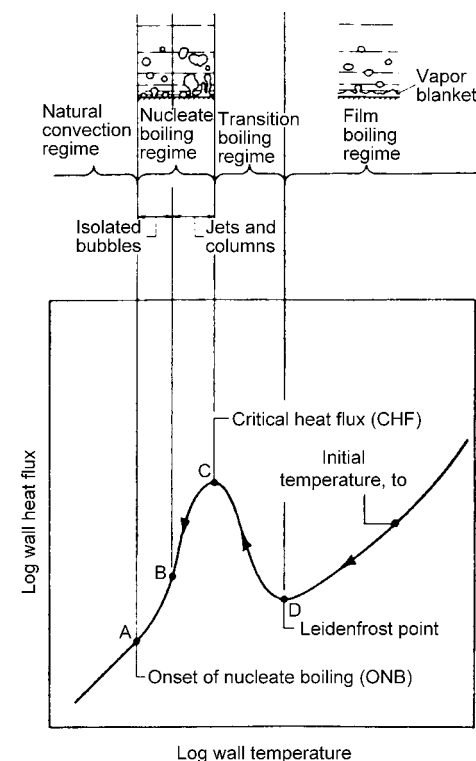


Fig. 3 Heat transfer versus surface temperature

24), showing the heat transfer from the part as a function of its surface temperature. Figure 3 also relates the surface temperature and heat-transfer coefficient to the mechanism of heat removal. Upon immersion, the part will first be surrounded by a vapor blanket, which will collapse as the part cools. The heat transfer through this vapor blanket is poor, and the part will cool slowly in this region.

The second region of the cooling curve is called the nucleate boiling region and corresponds to rapid heat transfer caused by direct contact of the part with the water. In this region, the part is still very hot and the water will boil vigorously. The high heat of water vaporization accounts for the very rapid heat transfer. In the third, or convective, cooling region, the surface of the part has cooled to a temperature below the boiling point of water. Only convective heat transfer occurs in this region.

The rate of heat removal from a quenched part can be increased by agitation, which reduces the stability of the vapor blanket that surrounds the part during the early stage of the quench. The effect of agitation on the cooling mechanism of a silver probe quenched by immersion in a 60 °C (140 °F) water bath is shown in Fig. 4 (Ref 6). A stream of water with a velocity of V_i was injected below the surface of the bath and directed at the probe. The greater the flow rate (agitation), the higher the temperature at which the vapor blanket mechanism was displaced by the much more effective nucleate boiling mechanism.

Cooling by the impact of a stream of droplets or liquid in spray quenching is also represented by Fig. 3. Fluid flow is used to accelerate the quench by rupturing the vapor blanket. The high agitation rates inherent in a spray quench also accelerate cooling in the nucleate boiling and convective cooling portions of the quench. Based on the mathematics needed to describe the spray systems, one author has stated that "spray cooling is also a sort of turbulent flow cooling" in all three regions of the cooling curve (Ref 25). In fact, it is useful to use the same terms to describe the heat-transfer mechanisms in both spray and immersion quenching.

Consider a series of water droplets moving toward the surface of a hot metal plate. The first droplet will contact the plate, and the surface temperature of the droplet at the point of contact will instantly rise. If the plate is sufficiently hot that the liquid in the contact zone is raised above its limiting superheat temperature (260 °C, or 500 °F, for water), then a portion of the liquid will evaporate in approximately 0.001 s (Ref 11). This evaporation will occur before the original spherical shape of the droplet has been significantly deformed. Because the contact region represents a small fraction of the mass of the drop, no significant heat removal from the plate will have occurred. The vapor film that forms will push the

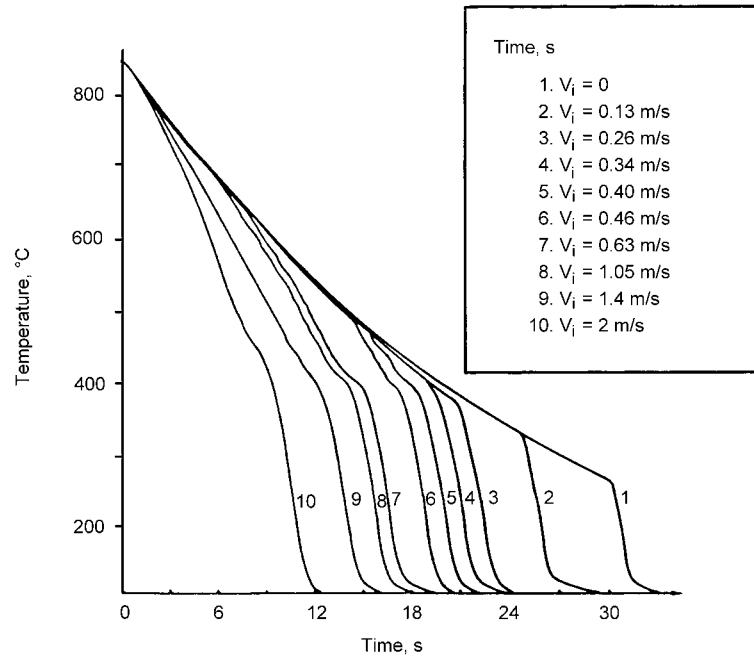


Fig. 4 Effect of agitation on the cooling curve

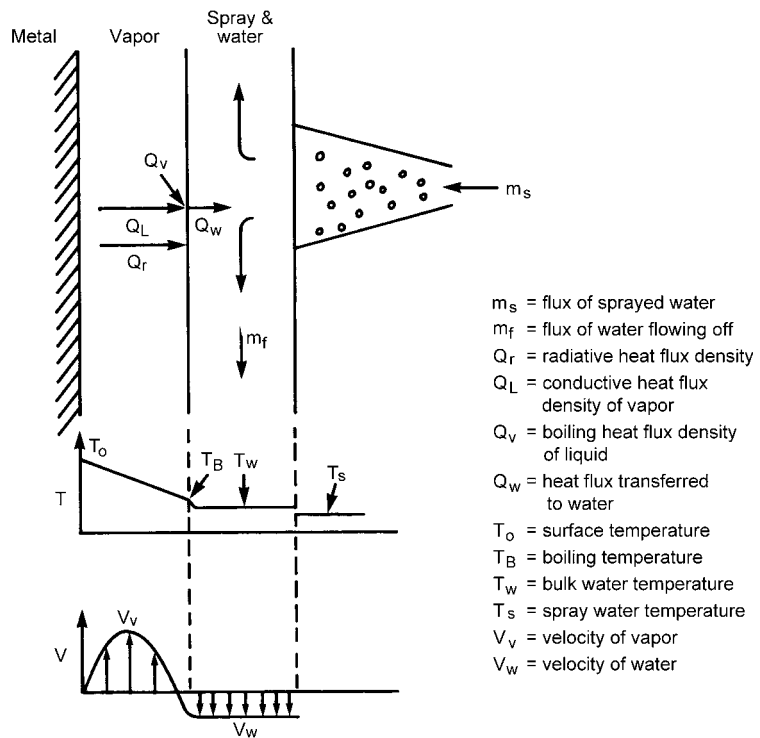


Fig. 5 Heat transfer in the vapor blanket portion of a spray quench

remaining portion of the droplet away from the plate. This vapor blanket provides a partial insulating barrier and inhibits the impact of additional droplets. Only those droplets with sufficient kinetic energy (momentum) will be able to penetrate the vapor film and strike the

plate. The result will be a film of liquid separated from the plate by a layer of vapor. This is called the film boiling or nonwetting phase.

The heat-transfer process for this state can be pictured as shown in Fig. 5 (Ref 2, 26). The heat-transfer rate through the vapor film is

relatively slow; therefore, there is a relatively slow decrease in the temperature of the plate. As the temperature falls, the thickness of the vapor film will decrease (Ref 5). At a characteristic temperature, which is dependent on the quench conditions, the droplets with higher-than-average kinetic energy will penetrate the vapor film, contact the plate, and begin to spread out on the surface. The temperature at which this occurs is called the Leidenfrost point. At this temperature, regions of the plate will be observed to be "wetted" by the quenchant.

The increased contact of the droplets with the metal surface will result in a corresponding increase in the heat transfer from the plate, causing the plate temperature to decrease more rapidly. A larger number of droplets will penetrate the film, and heat transfer will increase until the entire plate is wet with boiling liquid. At this point, the heat-transfer process is characterized by nucleate boiling. Further reduction of the plate temperature leads to an end of boiling, and the heat-transfer mechanism becomes dependent on convective cooling. In practice, the Leidenfrost temperature is found to be a function of the surface physical properties (Ref 6, 11, 27, 28) as well as its temperature (Ref 29, 30).

in the heat-transfer coefficient in the film and nucleate boiling regimes is required to abate such problems (Ref 31). The heat-transfer coefficient in water immersion quenching depends on the surface temperature of the part, the thermophysical properties of the part, and the initial water bulk temperature (Ref 32). Bamberger developed the following correlation of the heat-transfer coefficient in immersion quenching:

$$h_{\text{imm}} = \sqrt{K\rho C \exp\left(0.32 \frac{\theta_c - \theta_s}{\theta_b - \theta_c}\right)} + h_v + h_{\text{rad}} \quad (\text{Eq 1})$$

where:

$$h_{\text{rad}} = \sigma \varepsilon \frac{\theta_c^4 - \theta_a^4}{\theta_c - \theta_a}$$

and h_v is the stable film evaporation heat-transfer coefficient.

Investigating the heat transfer during quenching operations experimentally has many difficulties associated with it: measuring the temperature of the quenched part surface, determining the cooling rate, and quantifying the heat-transfer coefficient, especially with the two-phase nature of the flow. Recently, enormous attention has been given to modeling the heat-transfer coefficient in water immersion quenching, taking into consideration the boiling phase change. Srinivasan et al. (Ref 33, 34) developed a numerical code for immersion quenching processes. They used the most widely and commonly used relations to estimate the heat-transfer coefficient: Nusselt's method for film condensation to model film boiling, Zuber's relation for the critical heat flux, and Rohsenow's correlation for nucleate boiling. Their results matched the experimental data, except for some deviations in the case of complex shape quenching in the transition boiling regime.

The Effect of Quenching Agitation. When the part is first immersed in the water, the adjacent water layer evaporates rapidly, forming a vapor layer with low heat-transfer coefficient. Film boiling dominates until the surface

temperature falls to the Leidenfrost point or the vapor layer breaks at higher temperature. Agitation helps in breaking the vapor layer. The collapse starts from the corners of the part toward the center, augmenting the heat-transfer coefficient and increasing the cooling rate. Quenchant stirring, quenchant circulation, and submerged jet/spray mixing are the most commonly used techniques in quench tank agitation (Ref 35). Hernandez-Morales et al. (Ref 36) compared the heat-transfer rate in quenching of a steel disk in stagnant water and in agitated water. The stability of the vapor layer was severely affected by the water stirring, with an increase of 65% in the film boiling heat-transfer coefficient compared to stagnant water. Sedighi et al. (Ref 37) carried out an experimental investigation to quantify the agitation effect for flowing oil in the quench tank. They found that the time required to cool a cylinder from 860 to 100 °C (1580 to 212 °F) was reduced to half when the oil speed was quadrupled across the cylinder in the transverse configuration.

Spray Quenching

Spray quenching is superior to immersion quenching in that the cooling rate range can be adjusted by changing the flow rate and liquid pressure. Different spray and jet types can be used, as shown in Fig. 6. Spray quenching follows the boiling curve. The water droplet accelerates the quenching process by rupturing the vapor layer. The vapor layer is formed when water droplets from the spray touch the hot surface, and the liquid in contact with the plate evaporates, leaving the droplet in its spherical shape with a miniature contact with the surface. McGinnis and Holman (Ref 39) quantified the maximum heat flux from a single droplet on a heated surface. The maximum heat flux existed because of two opposing effects of the increasing temperature gradient and the weaker contact between the surface and the droplet with time. The vapor layer dramatically decreases heat removal from the surface. Because the

Water Quenching Heat Transfer

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Immersion Quenching

Oils and water are the most widely used liquids in steel quenching processes. Water, although most commonly used, causes cracks and distortion for quenched parts. Uniformity

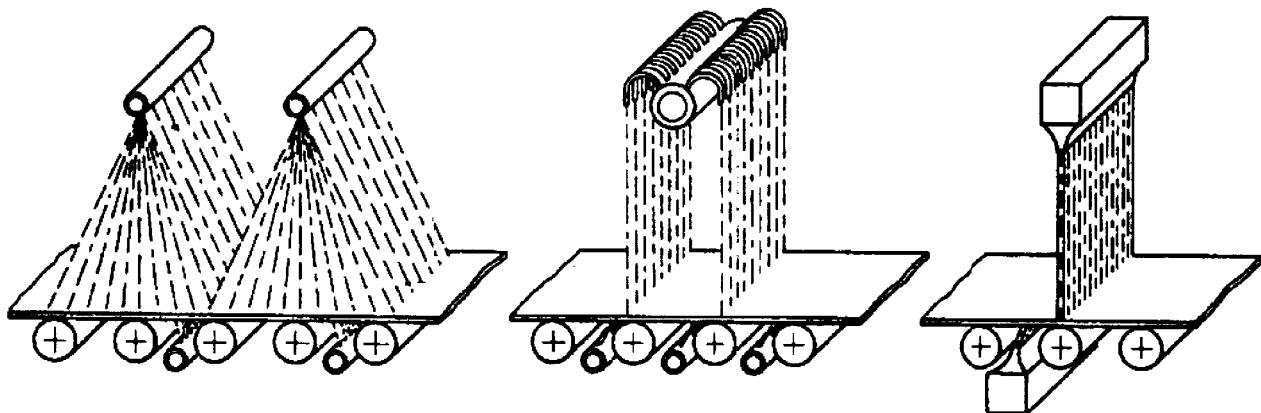


Fig. 6 The three main water-cooling configurations on runout tables. From left to right: spray, laminar, and water curtain. Source: Ref 38

droplets from the spray have high momentum, they break the vapor layer and wet the surface (Ref 35).

Sozbir et al. (Ref 40) compared low-mass-flux spray jets with air jets and found that the heat-transfer coefficient increases with the increase of the water-mass flux, between 0 and 7.67 kg/m²/s, for the mist-jet, as shown in Fig. 7. They pointed out that the increase in the heat flux was due to the fact that the droplet hits the same place on the surface. They indicated that the droplet velocity has a minor effect on the heat-transfer coefficient augmentation. High water velocity results in distortion of the droplet when it hits the surface.

Sengupta et al. (Ref 41) found that vapor film boiling in steel continuous casting is the dominant heat-transfer mode, and it is preferred because its heat-removal rate is mass-flux dependent. They established an empirical correlation for water heat-transfer coefficient with the following spray nozzle characteristics: type, nozzle separation, distance of the nozzle from the surface, and water-mass flux. Ciofalo et al. (Ref 42) carried out a transient study of the effect of different nozzles on swirl spray cooling. Their results are limited to the configurations they used. The reported heat flux is higher than the other reported heat fluxes for the same water-mass flux. Most of the empirical models developed for cooling steel plates cannot be applied for other conditions than the ones they have been developed for. Bamberger and Prinz (Ref 32) carried out experiments on spray cooling using water pressure between 0.12 and 0.5 MPa (0.017 and 0.07 ksi) and related the spray heat-transfer coefficient to the immersion heat-transfer coefficient given in Eq 1, as follows:

$$h_{spr} = 0.69 \log \frac{\dot{V}_w}{6 \times 10^{-4}} (h_{imm} - h_{rad}) + h_{rad} \quad (\text{Eq 2})$$

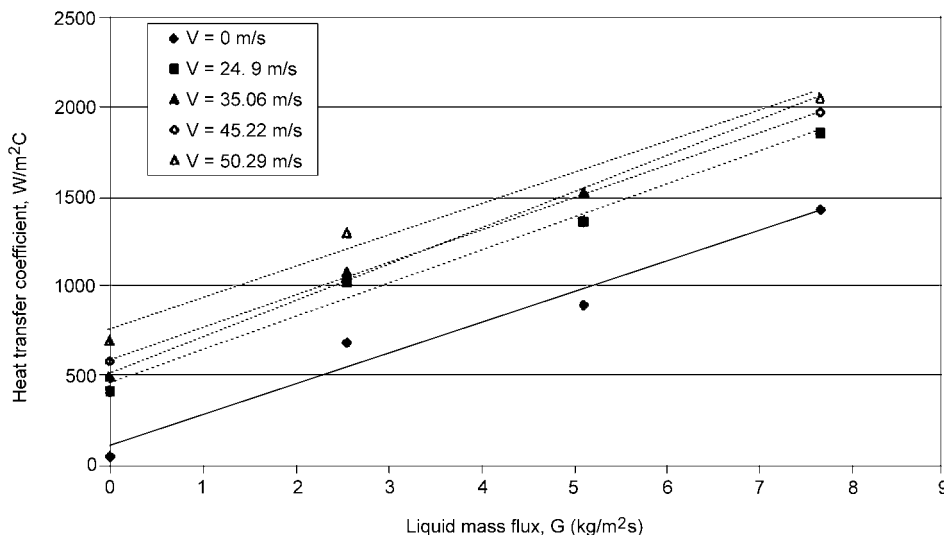


Fig. 7 Water-mass-flux effect on heat-transfer coefficient. Source: Ref 40

Mudawar and Valentine (Ref 13) carried out steady-state experiments up to 400 °C (750 °F) and were able to correlate the heat-transfer coefficient locally to the hydrodynamic properties of the spray (droplet velocity, 10.6 to 26.5 m/s, or 34.8 to 86.9 ft/s; droplet diameter, 0.434 to 2.005 mm, or 0.017 to 0.079 in.; and volumetric flux, 0.6 to 9.96 × 10⁻³ m³/s). The Nusselt number correlation in the single-phase region (Eq 3) is based on the Reynold's and Prandtl numbers. The Reynold's number is based on the characteristic velocity of the volumetric flux and the mean droplet diameter as the characteristic length. The heat-transfer rate in the nucleate boiling regime (Eq 4) is independent of the hydrodynamic properties and depends only on the surface and film temperatures:

$$Nu_{0.5} = 2.569 Re_{0.5}^{0.78} Pr_f^{0.56} \quad (\text{Eq 3})$$

$$q'' = 1.87 \times 10^{-5} (T_w - T_f)^{5.55} \quad (\text{Eq 4})$$

Jet Quenching

Because the water jet is not uniformly spread over the quenched part and it is more concentrated over a narrow region, there is hydrodynamic variations in the flow, which results in a nonuniform cooling pattern. However, impinging jets produce significantly high cooling rates. Several studies have been carried out investigating cooling rates in quenching processes using impinging jets.

Transient Studies. Ishigai et al. (Ref 43) obtained the boiling curves for a free impinging jet with jet velocity between 0.65 and 3.5 m/s (2.1 and 11.5 ft/s) and liquid subcooling of 5 to 55 °C (40 to 130 °F). They reported that the heat flux and the minimum surface

temperature are significantly affected by the degree of subcooling. At 2 m/s (6.6 ft/s) jet velocity, as the degree of subcooling increased from 5 to 15 °C (40 to 60 °F), the minimum surface temperature increased by more than 100 °C (180 °F). At higher degrees of subcooling, the heat flux tended to be constant with a decrease in the surface temperature. This phenomenon has been recently referred to as the shoulder heat flux (Ref 44), due to the intermittent surface rewetting.

Hall et al. (Ref 45) quenched a copper disk at 650 °C (1200 °F) with a water-jet velocity ranging from 2 to 4 m/s (6.6 to 13.1 ft/s). They noticed a sharp drop in the critical heat flux in the stagnation zone compared to the radial flow zone. They attributed that to the deceleration of the flow in the radial direction and the decrease in the shear force on the bubbles in the boundary layer, which led to a significant increase in the boundary layer thickness. The minimum heat flux was found to decrease with the radial position also, and the minimum temperature depended on the liquid film hydrodynamics. Mozumder et al. (Ref 46) reported similar trends. They also noted that the maximum heat flux position moves circumferentially outward following the wetting front. Karwa et al. (Ref 47) studied the wetting front propagation and hydrodynamic effects on a flat surface. Akmal et al. (Ref 48) studied the wetting front on cylindrical surfaces.

Islam et al. (Ref 49) captured different flow patterns while quenching a surface at 500 to 600 °C (930 to 1110 °F) with a 2 mm (0.08 in.) upward water jet of 5 to 80 °C (40 to 175 °F) of subcooling impinging on the surface with a velocity range from 3 to 15 m/s (9.8 to 49.2 ft/s). The six distinctive flow patterns depended on the surface thermal characteristics: material type, surface finish and aging, and surface temperature. They also studied the variations in the cooling rate due to the different boiling regimes.

Steady-State Investigations of Jet Cooling. Wolf et al. (Ref 50) carried out steady-state experiments using a 10 mm (0.4 in.) wide water jet kept at 50 °C (120 °F) of subcooling impinging on a horizontal flat surface. The surface heat flux was controlled by the means of direct heating using a direct-current power supply. The heat-transfer coefficient was quantified for the single-phase convection, partial, and fully nucleate boiling regimes. The effect of the jet velocity, between 2 and 5 m/s (6.6 and 16.4 ft/s) was clear in the single-phase and partial boiling regimes, where the flow hydrodynamics were dominant over bubble dynamics. The jet velocity had no significant effect in the fully developed boiling regime. The heat flux was not a function of the jet velocity and the only function of the surface superheat:

$$q'' = 63.7 (T_w - T_{sat})^{2.95} \quad (\text{Eq 5})$$

Robidou et al. (Ref 44, 51) were able to investigate the entire boiling curve at the jet

stagnation point and up to a location 55 times the jet width, which was 10 mm (0.4 in.). The maximum degree of subcooling was 20 °C (70 °F), and the maximum jet velocity was 1 m/s (3.3 ft/s). The jet impinged on a flat horizontal surface indirectly heated with ten heaters in an attempt to achieve constant surface temperature. They studied the effect of subcooling, jet velocity, immersion, and jet-to-heater spacing on the different boiling regimes. In the forced convection regime, the boiling curve was shifted up with the increase of subcooling and jet velocity. However, in the nucleate boiling regime, no parameter was found to have an effect on the heat flux and the boiling curve. The critical heat flux was found to increase with the increase in jet velocity and subcooling and to decrease with the distance from the stagnation point.

Modeling of jet cooling includes global and mechanistic forms.

Global Modeling. Most of the work cited in the previous two sections involved the development of an empirical correlation of the heat flux to the jet parameters and surface superheat. Although these kinds of relations are easy and fast to apply, they are limited to the conditions for which they have been developed. Omar et al. (Ref 52) developed an empirical/analytical model to predict the stagnation heat flux from

a free planar jet impinging on a horizontal flat surface. With the assumption of enough liquid superheat around the bubbles in the bubbly layer, bubbles departed the surface and then collapsed in the layer above, at the saturation temperature. Additional disturbances are induced due to bubble dynamics, which result in an enhancement of the heat transfer. Such enhancement has been represented by an additional diffusion term in the momentum and energy equations. The total diffusivity term in the momentum and energy equations (Eq 6 and 7) is assumed to be the sum of the molecular and the proposed bubble-induced diffusivity:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho_l} \frac{dP}{dx} + \frac{\partial}{\partial y} \left((\epsilon_m + \nu) \frac{\partial u}{\partial y} \right) \quad (\text{Eq 6})$$

$$u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} = -\frac{\partial}{\partial y} \left((\epsilon_h + \alpha) \frac{\partial T}{\partial y} \right) \quad (\text{Eq 7})$$

By solving the equations in a nondimensional form, they derived Eq 8 to calculate the nucleate boiling heat flux at the surface:

$$q''_{nb} = \epsilon^{+0.5} \rho_l C_p (T_{sat} - T_\infty) \sqrt{\frac{CV_{jv}}{w}} \frac{d\theta}{d\eta} \bigg|_{\eta=0} \quad (\text{Eq 8})$$

Solving Eq 8 requires a value for the diffusivity ϵ . They successfully correlated the

diffusivity to nondimensional groups that represent the flow:

$$\epsilon^+ = \frac{Re_b^{x_1} Ja_{sup}^{x_4} Ja_{sub}^{x_5}}{We_b^{x_2} + x_3} \quad (\text{Eq 9})$$

where x_1, x_2, x_3, x_4 , and x_5 are constants representing the effect of different forces on the bubble diffusivity. The model has been validated. It gave predictions of the heat flux with variations in the range of +30 and -15% from the experimental data.

Mechanistic Modeling of Jet Cooling. On the contrary to global models, mechanistic models have the ability to quantify total heat flux, and each component of heat flux independently, based on the submodels or experimental observations for bubble maximum diameter, release frequency, and density of the nucleation sites. One of the oldest wall flux partitioning models was proposed by Griffith (Ref 53) when he noticed a subcooled boiling region with low void fraction and a less subcooled boiling region with high void fraction. In the first region, the scattered static bubbles acted as surface disturbances, and the heat flux components are due to the single-phase flow and boiling, while in the second region the heat is assumed to transfer to the liquid through condensation of the bubbles.

Based on an earlier work of flow boiling heat transfer by Basu et al. (Ref 54), Omar (Ref 55) recently applied the wall heat flux partitioning concept to the stagnation zone heat flux produced by an impinging jet. Omar assumed that the total heat flux is the sum of three partitions due to forced convection, evaporation, and transient conduction heat fluxes, as shown in Fig. 8:

$$q''_w = q''_{FC} + q''_{TC} + q''_{ev} \quad (\text{Eq 10})$$

Omar found a closure for his mechanistic model by developing submodels for the bubble growth diameter and bubble growth terminal scenario. He assumed that after the bubble grows, it may slide or collapse in place, based on whether the dynamic equilibrium or the thermal equilibrium condition would be reached first (Fig. 9). Applying these equilibrium conditions, Omar calculated the maximum bubble

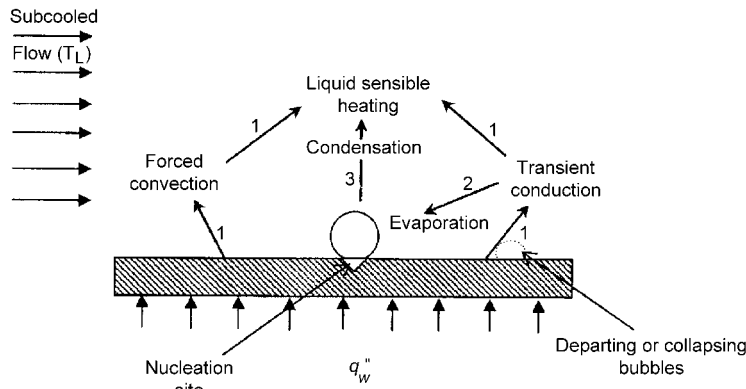


Fig. 8 Different heat flow paths. Source: Ref 55

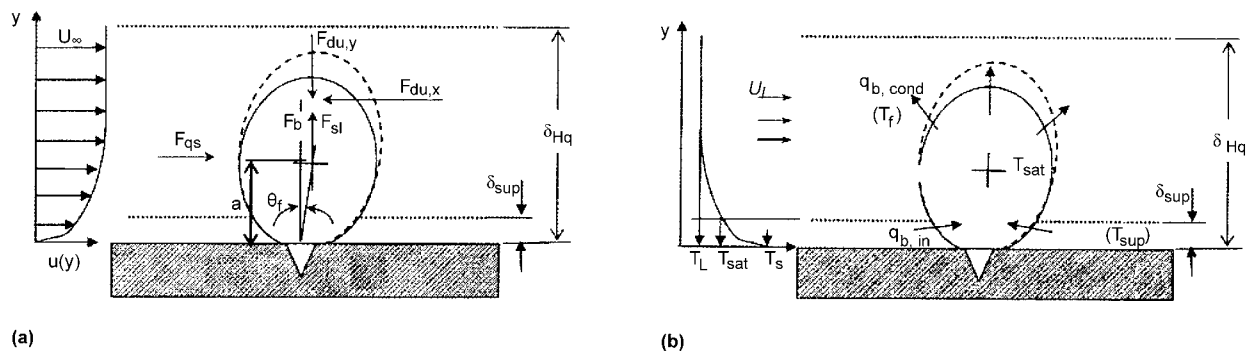


Fig. 9 Two bubble growth-termination scenarios. (a) Dynamic equilibrium. (b) Thermal equilibrium. Source: Ref 55

diameter. He used high-speed imaging and an intrusive optical probe to collect information about bubble dynamics (diameter, frequency, and number).

To find closure for his model, Omar also modeled the onset of nucleate boiling, bubble frequency, and nucleation-site density. Onset of nucleate boiling was correlated as a function of the film velocity, which is a function of the jet velocity. Using the optical probe, Omar measured the bubble frequency as a function of the dimensionless flow parameters. The model was validated using experimental data and was found to be $\pm 30\%$ accurate.

Summary

Water quenching is one of the most common ways in steel heat treatment to achieve the required hardness. Besides the enormous advantages of using water, it also serves a wide range of cooling rates in quenching. Each configuration has its difficulties in estimating the heat flux and the quenched part temperature, which requires more studies for predictiveness and expectedness of the process.

Spray quenching achieves more uniform surface cooling with no cracks and deformations, while immersion quenching is more suitable for complex large parts. Different agitation techniques are used to establish uniformity of the quenched part. Jet cooling has the capability of local cooling, which makes it, along with spray quenching, a good way to quench parts on a runout table.

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